

Developing a Distributed Trust Model for Distributed Energy Resources

N. Sonali Fernando, John M. Acken, Robert B. Bass

Department of Electrical and Computer Engineering

Portland State University

Portland, Oregon, USA

narmada@pdx.edu, acken@pdx.edu, robert.bass@pdx.edu

Abstract—Electric power system operators can manage distribution system reliability by coordinating end customer usage of distributed energy resources. The end customers in this regard are Service Provisioning Customers, who provide their energy resources to a Grid Service Provider, which in turn dispatches large aggregations of distributed energy resources to provide reliable service to the power system.

The security of this system relies upon information protection mechanisms, as described in IEEE 2030.5. However, in addition to preventive security measures, a monitoring function is required to ensure trustworthiness. Trust models are a method to detect and respond to both expected and unexpected behavior. Different trust models are required for various types and characteristics of each situation. This paper will describe the topics that must be considered when developing a trust model as it applies to distributed energy resources. The major contribution of this paper is the creation and application of a Distributed Trust Model applied to distributed energy resources.

Index Terms—Trust Model, Smart Grid Security, Energy Services Interface, Distributed Energy Resources, Energy Grid of Things

I. INTRODUCTION

The reliability of an electrical distribution system may be managed by intelligently adjusting end user demand. Such a system has been termed an “Energy Grid of Things” (EGoT). An EGoT uses aggregations of customer-owned distributed energy resources (DER) to provide essential reliability services through large-scale, coordinated dispatch of DER.

Because of its role in ensuring power system reliability, an EGoT should be considered critical infrastructure¹. And because an EGoT is dependent on internet communication, security is a significant concern. When devices are connected to the internet and are part of a network, security vulnerabilities increase. A Distributed

Trust Model (DTM) can help augment the security of a network by monitoring and evaluating all activities and notifying the appropriate parties when suspicious activities are identified. There are a variety of parameters and styles for creating DTMs. In this paper, we discuss how to pick a DTM for an EGoT system.

As a critical infrastructure, the security and trust of an EGoT system have a direct impact on national security. If an EGoT system becomes compromised, the reliability of an electrical power system could be jeopardized. Maintaining customer and infrastructure trust and security will result in gaining clients’ willingness to participate in an EGoT.

Our research group is developing a prototype EGoT system with two principle objectives: 1. Develop an Energy Service Interface (ESI), and 2. Develop a Distributed Trust model.

This paper will describe the topics that must be considered when developing a trust model as it applies to distributed energy resources. The major contribution of this paper is creation and application of Distributed Trust Model (DTM) within the context of the Energy Grid of Things (EGoT). Additionally, our DTM is compared with features and characteristics of current published trust models.

As defined by Widergren et al., “an ESI is a bi-directional, service-oriented, logical interface that supports the secure communication of information between entities inside and entities outside of a customer boundary to facilitate various energy interactions between electrical loads, storage, and generation within customer facilities and external entities.” [1]. An ESI provides a set of rules that govern the information exchange between a Grid Service Provider (GSP) and a Service Provisioning Customer (SPC) for purposes of facilitating the exchange of energy services. The GSP contracts with a Grid Operator (GO) to provide essential reliability services. The GSP then coordinates with SPCs for individual contributions to the requested services. Figure 1 shows an example of an electrical distribution system topology. Figure 2 shows the EGoT concept applied to

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¹Critical Infrastructure: “System and assets, whether physical or virtual, so vital to the U.S. that the incapacity or destruction of such systems and assets would have a debilitating impact on security, national economic security, national public health or safety, or any combination of those matters.”

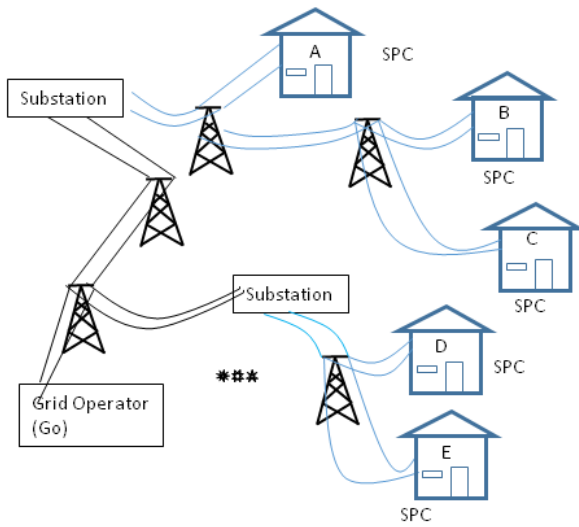


Fig. 1. An example Power Distribution system showing the Grid Operator and the customers.

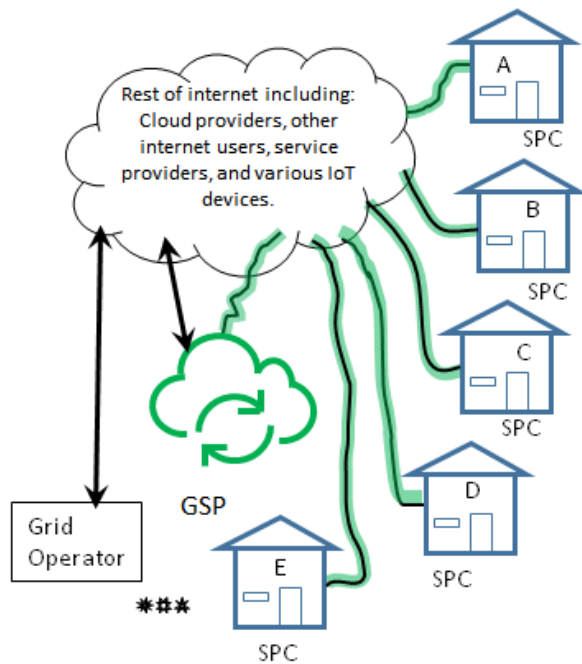


Fig. 2. An example of the Energy Grid of Things with the GO, GSP, and the SPC communication system.

that topology.

The EGoT will use the standard cybersecurity protocols as defined in IEEE 2030.5. IEEE 2030.5 is a standardized communication protocol that supports the exchange of energy services. Features of this protocol include the ability to discover DER, and to provide smart energy services such as demand response and flow

reservation.

The primary goal of the DTM is to monitor and to notify appropriate entities of any anomalies. The DTM evaluates the flow of data between a client and a server and evaluates the trust status of the sender and receiver. It then updates the trust and creates alerts and events as a response to the evaluation. The DTM augments existing cybersecurity measures without any interference to improve the trustworthiness of the system. The evaluation of the DTM is based on history, expectations, and specific events, etc. The DTM can distinguish a critical event versus a minor event and make decisions on when and whom to inform of such abnormalities. Another feature of the DTM is to keep track of the reputation of the device it is monitoring.

This paper discusses different features of the DTM that help in the selection of the most suitable DTM for a specific EGoT Architecture.

In section II, we discuss the different types of trust models and their characteristics. In section III, we present likely security attacks to the EGoT and attacks to the trust model. In section IV, we explain how self-trust or peer trust is established. In section V, we present how to evaluate security. Finally, we discuss how to strategically pick a DTM for a EGoT system based upon what is discussed in the paper.

II. DESCRIPTION OF TRUST MODEL TYPES

There are several characteristics of different types of trust models. These characteristics include organization, characterization, and components. This section provides a brief description of some examples of trust model characteristics.

A. Peer-to-Peer Trust Model

A Peer-to-Peer (P2P) trust model is a network of trust models wherein each communicates with other peer nodes. Some of the advantages of the P2P networks are improved network strength and flexibility, and variety in the available data. Disadvantages of a P2P network include a lack of accountability due to anonymity. This will lead nodes to be exposed to malicious peer ill treatment [2].

B. Hierarchical Trust Model

A Hierarchical Trust Model enables devices to be clustered in a tiered manner. This enables an upper tier to monitor known trusted clusters and use that learning to make a judgment on new clusters [3]. Communication between tiers to learn or alert about threats are also a possibility.

C. Centralized Trust Model

A Centralized Trust Model has one central node that is the main point of trust management. All the other nodes rely on that node for getting recommendations of trust.

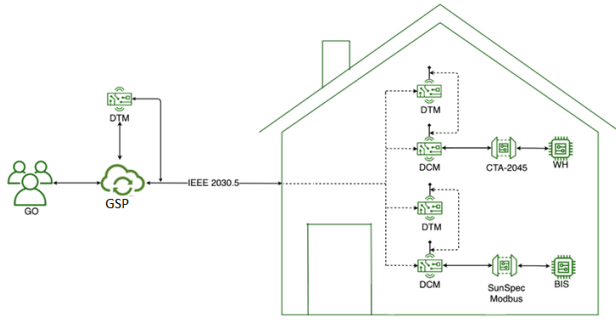


Fig. 3. Isolated Distributed Trust Model

D. Decentralized Trust Model

Since there is no central node, each node in a Decentralized trust model is responsible for evaluating and determining the trust of other nodes. An example of decentralization is self-certification which is a mechanism used by nodes for authenticating the senders identity [4].

E. Inner circle vs. outer circle

Some members are closely affiliated. They are considered inner circle. Some members are loosely affiliated. They are considered outer circle. Inner circle vs. outer circle or community-based trust evaluation can help identify a compromised node. For example, if one or more nodes provide a mistrust rating about a node in the inner circle, then it has a significant weight given that typically nodes that are part of the same circle provide positive feedback [5].

F. Isolated Distributed Trust Model

In Figure 3, the DTM is in an isolated location and is part of a Local Area Network (LAN). Its primary responsibility is to monitor directly connected devices for anomalies and report them to the appropriate parties.

G. Distributed vs. localized trust

Distributed trust can be applied to large scale networks where the communication exchange occurs among many peers and will be part of a Wide Area Network (WAN). For a localized trust the peers of the network are in a small area and may be part of a Local Area Network.

H. Trust Model components

Selecting and designing a trust model and the system that uses the trust models requires analyzing the specific application. Consideration must include storage limitations, organization of the trust model system, communication, and model equations.

The term “storage” refers to how trust is stored and what mechanism is used to access the trust metric when it is needed. Trust storage mechanisms can either be distributed or centralized. With a distributed mechanism,

there are multiple sensor nodes that are locally responsible for trust storage. In a centralized mechanism, a single main node is responsible for trust storage [6].

The trust score can be based on current and historic trust values. The trust metric varies based on the application. Trust metrics may have as few as one variable, or as many as 20 or more entries that represent total trust. In summary, anyone designing an architecture for DTM must decide which of the previously described organization, characteristics, and components will be used for their systems. We have picked F: Isolated DTM to evaluate locally.

From the above examples of trust model characteristics the best fitting trust model characteristics for the DTM for the EGoT would be as follow: For the DTM located at the SPC would have a P2P Decentralized trust model that is isolated. The DTM at the SPC would be hierarchically connected to the Central Distributed Trust Aggregator that is the centralized trust model. The the DTM architecture of the EGoT is recognized as a hybrid DTM due to the use of both Decentralized and Centralized trust model characteristics.

III. SECURITY ATTACKS

When designing a DTM architecture, the breadth of possible attacks must be considered, to the extent possible, so that the DTM is well-suited for detecting attacks within a specific context. In this section, we discuss both potential attacks on an EGoT system and possible attacks to the Distributed Trust Model itself.

A. Attacks on the Power Grid

Since the power grid is a critical infrastructure, attacks have severe ramifications. Across the nation there are multiple power grids that are interconnected. It is possible that an attack on one power grid may cascade onto the other power grids.

1) *Duplicate address selection/ detection*: Duplicated address selection is a form of attack to an EGoT system where the malicious source provides duplicate addresses of another node instead of its assigned address. This makes it difficult to detect the location/origin of the malicious node. Malicious nodes will be able to gain credibility by sending out addresses of a credible node.

2) *Man-in-the-Middle attack*: An EGoT system is vulnerable to man-in-the-middle attacks wherein a snooping party can watch and intercept information exchange between two parties. After observing and learning about the information exchange between two devices the intruder can pretend to be an authentic device. This will provide the intruder a means to execute attacks.

3) *Denial of Service*: With a denial-of-service attack, the attacker can block, alter, and obstruct incoming or outgoing messages. Within an EGoT system, communication exchange between energy consumers, energy

providers, and the distributor is critical for ensuring power system reliability. As such, it is critical to ensure communication channels are not disrupted.

4) *Eavesdropping*: Eavesdropping attackers tend to just listen in on the communication between two parties. Attackers may then use the learned information to understand how and what information is exchanged. Next, the attacker uses the gained knowledge to form an attack specific to the channel. Eavesdropping, or active reconnaissance, is a preliminary step for enabling many attacks, such as man-in-the-middle, denial-of-service, duplicate address selection.

B. Attacks on the Distributed Trust Model

Attacks to the Distributed Trust Model are also possible. It is easy to think that having an anomaly detection monitoring device like the DTM is sufficient for augmenting communication cybersecurity. What if the DTM itself is under attack? Then the system anomalies will go unnoticed until a disruptive attack occurs. It is best to assess potential DTM attacks in order to understand how to prevent such attacks.

1) *Badmouthing*: In a badmouthing attack, the DTM will report false information about the device it is monitoring. Examples of false information include announcing malicious nodes as good and non-malicious devices as malicious. This is the type of attack where the compromised DTM will report inaccurate information about devices in order to facilitate attacks on the greater network.

2) *On-off Attack*: In an on-off attack on the Trust Model, a node will occasionally behave abnormally or maliciously. In order to gain trust, the malicious node behaves normally the majority of the time. The malicious nodes can gain full trust, continuously collecting more information about the system, and drive occasional attacks with the possibility of limited consequences.

3) *Imposter Attack*: An imposter attack is defined as a malicious node that quickly switches its identity so that its behavior can go unnoticed. The malicious node can use an imposter identity to conduct the malicious attacks and then remove any imposter information that is used to carry out attacks.

4) *Conflicting behavior attack*: In this type of attack, the malicious node will act differently depending on the interacting node. To gain a higher reputation, a malicious node will behave appropriately with some nodes and inappropriately with other nodes. An example of this type of attack is when a node is providing good recommendations about the nodes in its immediate circle and provide bad recommendations of nodes outside the community.

5) *Sybil Attack*: In a Sybil attack, a malicious node creates multiple phantom nodes with imposter device IDs

to confuse and conduct attacks on the network [7]. A malicious node conducts an attack by creating phantom nodes which misleads the network and exhausts the resources of the network.

In contrast to attacks identified above, the purpose of our proposed DTM is to observe and raise alerts for GSP to provide information to detect if there are attacks or other anomalies.

IV. TRUST ESTABLISHMENT

There are many ways to establish trust. Going over different ways to learn about the credibility of another node will help establish and maintain trust. In this section, we discuss how one node is able to learn trust. We then present how one node is able to trust another node and why reputation is important to make the decision of trust or mistrust.

A. Self-Trust

For self-trust, the node evaluates its trust value to determine if it is malicious or not in addition to evaluating the trust of its peers. The ability for the Distributed Trust Model to calculate self-trust is an important trait to have [8].

B. Third-Party Trust

When a node is trying to contact another node in the network, it first asks the neighboring nodes for a trust recommendation before engaging in any communication. If the recommendation is positive, then the initial node will proceed with contacting the intended node. The polling of a recommendation from a neighbor is referred to as "third-party trust." A third-party trust is important in the Distributed Trust Model to increase accuracy in trust assessment.

C. Peer's Trust

A peer's trust is similar to a third-party trust. A Peer who is providing the recommendation must be directly connected to the peer it is recommending. This method helps the DTM gain behavioral information about that node from peers that regularly stay in touch.

D. Direct trust

Direct trust is when the peer can decide to trust or distrust based on past transactions. For some simplistic trust models, this is a common way of deciding to trust or not trust a directly-connected node.

E. Recommended trust

A Recommended trust is 'Peer A' asking 'Peer B' about the trustworthiness of 'Peer C'. This helps the asking peer to obtain a better understanding of the trustworthiness of a specific peer before contacting the

peer. This can be used to learn about trustworthiness of the peer who is providing the recommendations.

In comparison to these categories of trust our particular DTM is individual P2P DTM communicate with a Central Distributed Trust Aggregator (CDTA).

V. EVALUATING SECURITY

The evaluation of a specific security approach is based upon several metrics. One measure is operation performance. For real time protection of the physical power distribution, the timeliness of the response is essential, whereas creating a system for diagnosing failures after the fact has a bigger emphasis on correctness than on speed. In addition to measuring performance, the cost of the system must be considered. The cost to implement a security feature must be evaluated relative to the cost of a security failure or the resources available.

For any evaluation system, there is a standard method called hypothesis testing. As this is applied to trust model for an EGoT system, we define the hypothesis (h_0) to test whether an entity is trustworthy. Figure 4 shows the choices of a trust model evaluation versus the actual trustworthiness of an entity. When the model predicts an entity is trustworthy, this is a Correct Trust Evaluation (CT). However, when the model predicts that an entity should not be trusted and in fact the entity is not trustworthy this is a Correct Distrust evaluation (CD). Both are measure of the trust model working. However, a type I error occurs when the model evaluation indicates trustworthiness, but the entity is not trustworthy. This we call Mistaken Trust (MT). Alternatively, if the model evaluates that an entity should not be trusted that is in fact trustworthy, this is the case of Mistaken Distrust (MD). Any model will have a trade-off between the accuracy of missing an untrustworthy case versus falsely suspecting a trustworthy entity. The trade-off varies with the strictness of testing whether an entity is trustworthy. Figure 5 shows the effects of changing the strictness threshold on the error rates of Mistaken Trust versus Mistaken Distrust.

VI. SUMMARY AND CONCLUSION

An electric power system may be made to be more reliable through the implementation of an EGoT system, wherein customer-owned DER are aggregated and dispatched in order to provide essential reliability services. Because the electric power system is an integral part of a country's critical infrastructure, security is essential. Information security may be protected by applying cybersecurity measures. However, no security system can predict all future possible attacks. A trust-based monitoring approach can be added to augment traditional cybersecurity mechanisms. One such approach is to

	Entity is Trustworthy	Entity is NOT Trustworthy
Evaluated As Trustworthy	CT Correct Trust Evaluation	MT Mistaken Trust
Evaluated As NOT Trustworthy	MD Mistaken Distrust	CD Correct Distrust Evaluation

Fig. 4. Table showing Trust evaluation with respect to actual trustworthiness.

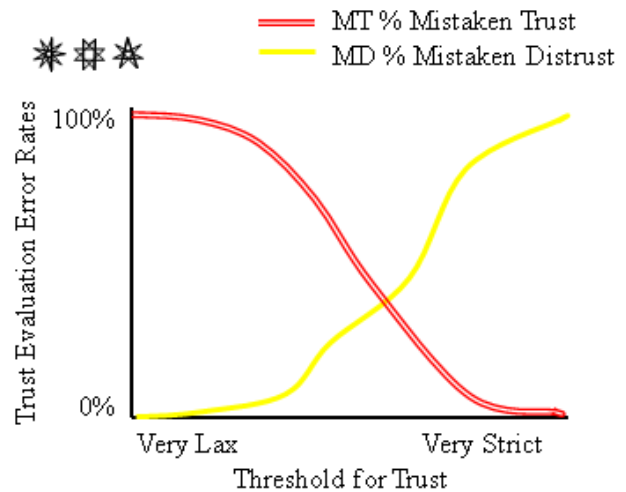


Fig. 5. Trade-off between setting evaluation threshold and error rates.

implement trust models. We presented the considerations that should be made when selecting and designing a trust model system, with an emphasis on applying the trust model to an EGoT system. We described a sample of various trust models and organizations. Then we described some potential security attacks, against both the system and the trust model itself. Finally, we described several of the metrics for evaluation of trust, with an emphasis on the error rates.

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